Randomized average-case easiness yields subexponential simulations of promise polynomial hierarchy levels

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Editorial scope. Counted original Theorem43 once: each promised polynomial-hierarchy level has a randomized subexponential simulation under the exact stated higher-level randomized average-case easiness assumption. Complete Theorem40 linear witness compression and all source-local pKt/rKt/PRG support included, not separately counted. No unconditional simulation, deterministic algorithm or collapse of PH to BPP claimed.

Allowed standard prerequisites (editorial).

- Standard polynomial hierarchy/promise classes and Lautemann simulation; imported DPG reconstruction [11] and pseudodeterministic PRG hardness ingredients [19,39,35]. Exact imported PRG/circuit-hardness statements retained. All source-local padding, reconstruction, symmetry, oracle elimination, witness compression and hierarchy induction proofs included.

2 Preliminaries (necessary author context)

2.1 Notation

We say that a function $p : \mathbb{N} \to \mathbb{N}$ is non-decreasing if $p(\ell) \geq \ell$ for every integer $\ell \geq 0$. Throughout the paper, we will rely on constructions whose associated complexity parameters (e.g., running time) can be bounded by some polynomial p. We will implicitly assume that the polynomials provided in these results are non decreasing in order to avoid including this condition in every statement. This can be done without loss of generality by the monotonicity of our bounds. For a probability distribution D_n, we use $\mathrm{supp}(D_n)$ to denote its support. Given an element x, we let $D_n(x)$ denote the probability of x under D_n. We denote by U_n the uniform distribution over n-bit strings. Let D_1 and D_2 be probability distributions supported over a set X, $A : X \to \{0,1\}$, and $\varepsilon \geq 0$. We say that A ε-distinguishes D_1 and D_2 if

$$\left| \Pr_{x \sim D_1}[A(x) = 1] - \Pr_{x \sim D_2}[A(x) = 1] \right| \geq \varepsilon.$$

If A is a randomized algorithm, each probability in the expression above also takes into account the internal randomness of A, which will be denoted by r_A.

2.2 Average-case complexity

Recall that a pair $(L, \mathcal{D})$ is a distributional problem if $L \subseteq \{0,1\}^*$ and $\mathcal{D} = \{D_n\}_{n\geq 1}$ is an ensemble of probability distributions, where each D_n is supported over $\{0,1\}^*$. Let $\mathcal{D} = \{D_n\}_{n\geq 1}$ be an ensemble of distributions. We say that $\mathcal{D} \in$ PSamp if there is a randomized polytime algorithm A such that, for every $n \geq 1$, $A(1^n)$ is distributed according to D_n. More generally, we use $D_n \in$ Samp$[T(n)]/a(n)$ to denote that D_n can be sampled by an algorithm that runs in time $T(n)$ and has advice complexity $a(n)$. We let DistNP denote the set of distributional problems $(L, \mathcal{D})$ with $L \in$ NP and $\mathcal{D} \in$ PSamp.

Definition 4 (Avg$_\delta$BPP **[3]).** Let $(L, \mathcal{D})$ be a distributional problem, and $\delta : \mathbb{N} \to [0,1]$. We say that $(L, \mathcal{D}) \in$ Avg$_\delta$BPP if there is randomized polytime algorithm A such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n) = \perp] \geq 1/4\right] \leq \delta(n),$$

where r_A denotes the internal randomness of A. Such an algorithm A is called an randomized errorless heuristic for $(L, \mathcal{D})$ with failure probability at most δ.

Definition 5 (AvgBPP [3]). [1] We say that $(L, \mathcal{D})$ is in AvgBPP if there exist a randomized algorithm A and a polynomial p such that

1. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n,\delta) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n, \delta > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n,\delta) = \perp] \geq 1/4\right] \leq \delta(n),$$
3. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$, $A(x;n,\delta)$ runs in time at most $p(n/\delta)$.

Such an algorithm A is called a randomized errorless heuristic scheme for $(L, \mathcal{D})$.

Lemma 6. Let $(L, \mathcal{D})$ be a distributional problem in Avg$_{n^{-1}}$BPP. There exists a randomized polynomial-time algorithm B such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$, $B(x;n) \in \{0,1\}$,
2. for every $n > 0$, and every $x \in \text{supp}(D_n)$, if $x \in L$, then $\Pr_{r_B}[B(x;n) = 1] \geq 1 - 1/n$, and
3. for every $n > 0$, $\Pr_{x\sim D_n, r_B}[B(x;n) = L(x)] \geq 1 - 3/n$.

[1] Some authors define AvgBPP as $\bigcap_{c>0}$ Avg$_{n^{-c}}$BPP. All our results also hold with respect to that definition.

Proof. Let A be an $\mathrm{Avg}_{n^{-1}}\mathrm{BPP}$ algorithm for $(L, \mathcal{D})$ as in Definition 4. Let A' be the algorithm obtained, using standard amplification techniques,[2] such that for all $x \in \mathrm{supp}(D_n)$,

$$\Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n \tag{1}$$

and

$$\Pr_{x \sim D_n}\left[\Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\right] \leq 1/n. \tag{2}$$

Let B be the randomized algorithm which, on input $x \in \mathrm{supp}(D_n)$, simulates $A'(x;n)$ and outputs 1 if $A'(x;n) \in \{1, \perp\}$ or 0 if $A'(x;n) = 0$. By Equation (1), if $x \in L$, then

$$\Pr_{r_B}[B(x;n) = 0] = \Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n.$$

Let $\mathrm{bad} := \{x \in \mathrm{supp}(D_n) \mid \Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\}$. By Equation (2), at most a $(1/n)$-measure of $x \in \mathrm{supp}(D_n)$ are in bad. Thus,

$$\begin{aligned}
\Pr_{x \sim D_n, r_B}[B(x;n) \neq L(x)] &\leq \Pr_{x, r_{A'}}[A'(x;n) = \perp] + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq \Pr_{x}[x \in \mathrm{bad}] + \Pr_{x, r_{A'}}[A'(x;n) = \perp \mid x \notin \mathrm{bad}] \\
&\quad + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq 1/n + 1/n + 1/n = 3/n.
\end{aligned}$$

This completes the proof. ◁

2.4 Direct product generators (DPGs)

Definition 13 (Direct Product Generator (DPG) [11, Definiton 3.10]). For $k, n \in \mathbb{N}$, we define the k-wise direct product generator to be the function

$$\mathrm{DP}_k : \{0,1\}^n \times \{0,1\}^{nk} \to \{0,1\}^{nk+k}$$

such that

$$\mathrm{DP}_k(x; z^1, \ldots, z^n) := (z^1, \ldots, z^k, x \cdot z^1, \ldots, x \cdot z^k).$$

Lemma 14 (DPG Reconstruction [11, Lemma 3.14]). For any $n, k \in \mathbb{N}$ with $k \leq 2n$, and $\varepsilon > 0$, there exists a pair of algorithms A and $\mathrm{Recon}^{(-)}$ such that

- Recon^{D} takes oracle access to an oracle $D : \{0,1\}^{nk+k} \to \{0,1\}$.
- $A : \{0,1\}^n \times \{0,1\}^r \to \{0,1\}^k$ is called the advice function and is computable in time $\mathrm{poly}(n/\varepsilon)$.
- $\mathrm{Recon}^{(-)} : \{0,1\}^k \times \{0,1\}^r \to \{0,1\}^n$ is called a reconstruction procedure and is computable in time $\mathrm{poly}(n/\varepsilon)$.[3]

[2] In more detail, A' can be obtained from A by running it $O(n)$ times and selecting the most common output.

[3] We note that if the oracle is a uniform algorithm that runs in time $t(m)$ on inputs of length m, the reconstruction procedure can be computed in time $t(nk+k) \cdot \mathrm{poly}(n/\varepsilon)$.

- The randomness complexity r is at most $\text{poly}(n/\varepsilon)$.
- For any string $x \in \{0,1\}^n$ and any function D that ε-distinguishes $\text{DP}_k(x; U_{nk})$ from U_{nk+k}, it holds that

$$\Pr_{w \sim \{0,1\}^r}[\text{Recon}^D(A(x,w),w) = x] \geq 1/\text{poly}(n/\varepsilon).$$

2.7 Encoding tuples (author context)

We will also need to consider distributions that generate input instances that are tuples, while the only known information to the distributions is the length of the instances. To do this, we will encode the format as the length of the instances. More specifically, we use the following way to encode tuples [3].

Proposition 16 ([3]; see also [6, Proposition 3.1]). There is a pair of polytime encoding-decoding algorithms $\text{Enc} : \mathbb{N}^* \to \mathbb{N}$ and $\text{Dec} : \mathbb{N} \to \mathbb{N}^* \cup \{\perp\}$ such that

1. For $k \in \mathbb{N}$ and k integers $n_1, n_2, \ldots, n_k$, we write $\langle n_1, n_2, \ldots, n_k \rangle := \text{Enc}(n_1, n_2, \ldots, n_k)$, and we have
   (a) $\sum_{i=1}^{k}(2 \cdot (\lfloor \log n_i \rfloor + 2) + n_i) \leq \langle n_1, \ldots, n_k \rangle \leq O_k(\prod_{i=1}^{k}(n_i \cdot \log^2 n_i))$.
   (b) $\text{Dec}(\langle n_1, n_2, \ldots, n_k \rangle) = n_1, \ldots, n_k$. Here, Dec takes the binary representation of an integer and outputs k integers which are also represented in binary.
2. $\text{Dec}(u) = \perp$ if $u \neq \langle \vec{n} \rangle$ for any $\vec{n} \in \mathbb{N}^*$.

Proof. Given k integers $n_1, n_2, \ldots, n_k$, the encoding algorithm Enc first obtains the binary string

$$z := \alpha_1 \circ 01 \circ \alpha_2 \circ 01 \circ \alpha_k \circ 01 \circ \text{bin}(n_1) \circ \text{bin}(n_2) \circ \cdots \circ \text{bin}(n_k),$$

where for each $i \in [k]$, α_i is obtained from $\text{bin}(|\,\text{bin}(n_i)|)$ with every bit doubled. Then Enc outputs the integer n whose binary representation is z. It is easy to see that given the binary representation of n, which is simply the string z, one can recover the $n_1, n_2, \ldots, n_k$ in time polynomial in $|z|$. Also, item (a) can be verified by a simple calculation. ◁

3 Probabilistic Time-Bounded Kolmogorov Complexity (author excerpts)

In this section, we formally define probabilistic Kolmogorov complexity and prove some useful properties of this notion, which will be used later in proving our main results.

Definition 17 (pK^t). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N}$, an oracle A, and $\delta \in [0,1]$, the δ-probabilistic A-oracle t-bounded Kolmogorov complexity of x given y is defined as

$$\mathrm{pK}^{t,A}_\delta(x \mid y) := \min\left\{k \;\middle|\; \Pr_{w\sim\{0,1\}^t}[\exists M \in \{0,1\}^k, M^A(w,y) \text{ outputs } x \text{ within } t \text{ steps}] \geq \delta\right\},$$

where M is a RAM-machine.[4] We omit the subscript δ if $\delta = 2/3$, omit the superscript A if $A = \emptyset$, omit "$\mid y$" if y is the empty string, and omit the superscript t if $t = \infty$. For simplicity, in the rest of this section we only consider pK^t without oracles. It is easy to see that all the results also hold with any oracle.

Lemma 21 (Success Amplification). For any string $x \in \{0,1\}^n$, time bound $t \in \mathbb{N}$, and $0 \leq \alpha < \beta \leq 1$, we have

$$\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq \mathrm{pK}^t_\alpha + O(\log(q/\alpha)),$$

where $q = \ln(1/(1-\beta))$.

Proof. Suppose $\mathrm{pK}^t_\alpha(x) = k$. Then by definition, we have

$$\Pr_{w\sim\{0,1\}^t}[\exists M_w \in \{0,1\}^k \text{ such that } M_w(w) \text{ outputs } x \text{ within } t \text{ steps}] \geq \alpha.$$

Call such a w good. After sampling ℓ independent $w^{(1)}, w^{(2)}, \ldots, w^{(\ell)} \in \{0,1\}^t$, the probability that no $w^{(i)}$ is good is at most $(1-\alpha)^\ell \leq e^{-\alpha\ell}$, which can be made at most $1-\beta$ by choosing $\ell = q/\alpha$. It follows that, with probability at least β over a random $w = (w^{(1)}, w^{(2)}, \ldots, w^{(q/\alpha)}) \in \{0,1\}^{qt/\alpha}$, there exists an index $1 \leq i \leq q/\alpha$ (described with at most $\lceil \log(q/\alpha) \rceil$ bits) and a program $M_{w^{(i)}} \in \{0,1\}^k$ such that $M_{w^{(i)}}(w^{(i)})$ outputs x within $O(qt/\alpha)$ steps. Hence, $\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq k + O(\log(q/\alpha))$. ◁ Note that the parameter α affects the time complexity in the resulting $\mathrm{pK}^{O(qt/\alpha)}_\beta$ bound. Lemma 21 is sufficient in applications where $\alpha \geq 1/\operatorname{poly}(n)$.

3.2 Bounding pK^t via DPG reconstruction (selected original case)

The following is a key lemma for pK^t that we will use in proving our main results related to randomized average-case complexity.

Lemma 22 (pK^t Reconstruction Lemma). For $\varepsilon > 0$, $x \in \{0,1\}^n$, $s \in \mathbb{N}$, and $k \in \mathbb{N}$ satisfying $k \leq 2n$, let D be a randomized algorithm that takes an advice string β and runs in time t_D such that D ε-distinguishes $\mathrm{DP}_k(x; U_{nk})$ from U_{nk+k}. Then there is a polynomial p_{DP} such that

$$\mathrm{pK}^{\widetilde{O}(t_D)\cdot p_{\mathrm{DP}}(n/\varepsilon)}(x \mid \beta) \leq k + \log p_{\mathrm{DP}}(nt_D/\varepsilon).$$

[4] We use the RAM model in this definition for convenience: it simplifies the time bound estimates for a machine M that takes as input both randomness w and an "auxiliary" string y, and allows both w and y to be as long as the running time of the RAM-machine M; cf. [29] for a similar definition in the case of deterministic time-bounded conditional Kolmogorov complexity. Alternatively, we could use a TM model, with inputs w and y presented on two different tapes. However, we prefer the RAM model as it's also useful in the context of fine-grained worst-case to average-case reductions [6].

Proof. We will view the distinguisher D as a deterministic algorithm that takes a string of t_D bits, denoted by r_D, as its internal randomness. By our assumption (and dropping the absolute value sign without loss of generality), we have

$$\mathbb{E}_{y\sim\{0,1\}^{nk},r_D\sim\{0,1\}^{t_D}}[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_{z\sim\{0,1\}^{nk+k},r_D\sim\{0,1\}^{t_D}}[D(z;r_D)]\geq\varepsilon.$$

Then by an averaging argument, we have

$$\Pr_{r_D}[\mathbb{E}_y[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_z[D(z;r_D)]\geq\varepsilon/2]\geq\varepsilon/2. \tag{4}$$

In other words, with probability at least $\varepsilon/2$ over the internal randomness r_D, $D(-,r_D)$ is an $(\varepsilon/2)$-distinguisher for $\mathrm{DP}_k(x;U_{nk})$ and U_{nk+k}. Let us say that r_D is good if this is true. Let $\mathrm{Recon}^{(-)}:\{0,1\}^k\times\{0,1\}^r\to\{0,1\}^n$ be the reconstruction procedure in Lemma 14 that works for distinguishing parameter $\varepsilon/2$, where the randomness complexity r is $\mathrm{poly}(n/\varepsilon)$. We have

$$\begin{aligned}
&\Pr_{w\sim\{0,1\}^r,r_D\sim\{0,1\}^{t_D}}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x]\\
&\geq \Pr_{w,r_D}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x \mid r_D \text{ is good}]\cdot\Pr_{r_D}[r_D \text{ is good}]\\
&\geq \frac{1}{\mathrm{poly}(n/\varepsilon)}\cdot(\varepsilon/2). \qquad \text{(Lemma 14 and Equation (4))}
\end{aligned}$$

The above implies that for at least $1/\mathrm{poly}(n/\varepsilon)$ fraction of the randomness (w,r_D), there exists a program M, which takes k bits for some α (that can depend on the randomness) and the advice of β used by D, such that M simulates $\mathrm{Recon}^{D(-,r_D)}$ on (α,w) and outputs x. Note that since Recon^D runs in time $t_D\cdot\mathrm{poly}(n/\varepsilon)$, our program M can be made to run in time $T:=\widetilde{O}(t_D)\cdot\mathrm{poly}(n/\varepsilon)$. Therefore, we have

$$\mathrm{pK}^T_{1/\mathrm{poly}(n/\varepsilon)}(x\mid\beta)\leq k+O(\log(nt_D/\varepsilon)).$$

Then the lemma follows easily from Lemma 21. ◁

3.4 Symmetry of information (original Item1 and its full proof)

Lemma 26 (Symmetry of information for pK^t and rK^t under the average-case easiness of NP), Item 1. If DistNP $\subseteq$ AvgBPP, then there exist polynomials p and p_0 such that for all sufficiently large $n,m\in\mathbb{N}$, all $t\geq p_0(n,m)$, and all $0\leq\tau\leq t$ the following holds: for every $x\in\{0,1\}^n$ and every family $\{y_r\in\{0,1\}^m\mid r\in\{0,1\}^\tau\}$, with probability at least $9/10$ over $r\in\{0,1\}^\tau$,

$$\mathrm{pK}^t(x,y_r\mid r)>\mathrm{pK}^{p(t)}(x\mid r)+\mathrm{pK}^{p(t)}(y_r\mid x,r)-\log p(t).$$

Moreover, the special case without conditioning on r also holds for both items above, i.e., when r is the empty string (for $\tau=0$) and y is an arbitrary single string.

Proof of Item 1. Define a language

$$L := \{(u, v, w, w', 1^s) \mid \exists M \in \{0,1\}^s, M(w, w') \text{ prints } uv \text{ in } |w| \text{ steps, and } s = |u|+|v|-10\}.$$

Note that $L \in$ NP. Define a distribution family $\mathcal{D} = \{D_{\langle nk+k, mk'+k', 2t, \tau, s\rangle}\}$ as follows: sample $u \sim U_{nk+k}$, $v \sim U_{mk'+k'}$, $w \sim U_{2t}$, and $w' \sim U_\tau$, and then output $(u, v, w, w', 1^s)$. Note that the ensemble $\mathcal{D}$ is in PSamp under an appropriate encoding of its defining tuple (for simplicity, we omit the padding in the output of $\mathcal{D}$; see Section 2.1). Using the assumption that DistNP $\subseteq$ AvgBPP, it follows that $(L, \mathcal{D}) \in$ AvgBPP. Let B be a randomized algorithm for $(L, \mathcal{D})$ as in Lemma 6. Let $x \in \{0,1\}^n$ and $\{y_r\}_r \subseteq \{0,1\}^m$ be given (for sufficiently large n and m). Let $k, k' \in \mathbb{N}$ be arbitrary parameters such that $k \leq n$ and $k' \leq m$. Observe that there exists a polynomial p_0 such that for any $t \geq p_0(n, m)$, some constant d, and all $z \in \{0,1\}^{nk}$, $z' \in \{0,1\}^{mk'}$, and $r \in \{0,1\}^\tau$,

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) \leq \mathrm{pK}^t(x, y_r \mid r) + |z| + |z'| + d \log t. \tag{6}$$

Here $p_0(n, m)$ reflects the time required to deterministically compute $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$ given x, y_r, z, z', and $d \log t$ bits of information to delineate x from y_r. Let $t \geq p_0(n, m)$. By a counting argument, for independently random u, v, w, and w',

$$\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \leq \frac{2^s \cdot 2^{|w|+|w'|}}{2^{|u|+|v|+|w|+|w'|}} = 2^{s-|u|-|v|} = 2^{-10},$$

where the last equality is by the definition of L requiring that $s = |u| + |v| - 10$. Then

$$\begin{aligned}\Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) = 1] \leq &\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \\ &+ \Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) \neq L(u, v, w, w', 1^s)] \\ &\leq 2^{-10} + (3/n). \end{aligned} \tag{7}$$

By Markov's inequality, for at least 9/10 fraction of random strings $r \in \{0,1\}^\tau$, we get

$$\Pr_{u,v,w,r_B}[B(u, v, w, r, 1^s) = 1] \leq 10 \cdot (2^{-10} + (3/n)) \leq 1/10. \tag{8}$$

Fix any such r so that Equation (8) holds. This also fixes the string y_r. Next we show, using a hybrid argument, that $B(-, U_{2t}, r, 1^s)$ cannot distinguish between the uniform distribution and the distribution $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$, for random independent z, z', where $k \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(x \mid r)$ and $k' \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(y_r \mid x, r)$. This will imply that

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] < 2/3$$

for some z, z', yielding the desired lower bound on $\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r)$ via our choice of s and the definition of L. We give the details of the hybrid argument next. First, toward a contradiction, suppose

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] > 1/2.$$

In this case, comparing with Equation (8), we get a randomized distinguisher (with advice) for $\mathrm{DP}_k(x; U_{nk})$, defined by sampling $v \sim U_{mk'+k'}$ and $w \sim U_{2t}$, and outputting $B(-, v, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q(t)}(x \mid r) \leq k + \log q(t) \tag{9}$$

for some polynomial q such that $q(t) \geq p_{\mathrm{DP}}(t_B \cdot 3 \cdot n)$ whenever $t \geq n + m$ and $s \leq n^3$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, v, w, r, 1^s)$. Define $k := \mathrm{pK}^{q(t)}(x \mid r) - \log q(t) - 1$ so that Equation (9) does not hold. Assume for now that $k > 0$. Hence,

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] \leq 1/2. \tag{10}$$

Again, toward a contradiction, suppose that for all z, z',

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \geq 2/3.$$

By definition of B, this implies that

$$\Pr_{z,z',w,r_B}[B(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) = 1] \geq (2/3)(1 - 1/n) > 5/8.$$

In this case, comparing with Equation (10), we get a randomized distinguisher (with advice) B' for $\mathrm{DP}_{k'}(y_r; U_{mk'})$, defined by sampling $z \sim U_{nk}$, $w \sim U_{2t}$, and outputting $B(\mathrm{DP}_k(x; z), -, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q'(t)}(y_r \mid x, r) \leq k' + \log q'(t) \tag{11}$$

for some polynomial q' with $q'(t) \geq p_{\mathrm{DP}}(t_{B'} \cdot 8 \cdot m)$ whenever $t \geq n + m$ and $s \leq n^3$, where $t_{B'}$ denotes the time required to compute B'. We now choose $k' := \mathrm{pK}^{q'(t)}(y_r \mid x, r) - \log q'(t) - 1$ so that Equation (11) does not hold. Assume for now that $k' > 0$. Hence, there exist z and z' such that

$$\begin{aligned}&\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \\ &= \Pr_w[\exists M \in \{0,1\}^s, M(w, r) \text{ outputs } (\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z')) \text{ within } |w| \text{ steps}] < 2/3,\end{aligned}$$

which implies that

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > s.$$

For this choice of z, z', by definition of s we get that $s = |u| + |v| - 10 = |z| + k + |z'| + k' - 10$, and so

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > |z| + k + |z'| + k' - 10.$$

Combining this inequality with Equation (6), we get

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) &\geq \mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) - |z| - |z'| - d \log t \\ &> k + k' - d \log t - 10.\end{aligned} \tag{12}$$

The definitions of k and k' then imply that

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{q(t)}(x \mid r) + \mathrm{pK}^{q'(t)}(y_r \mid x, r) \\ - \log q(t) - \log q'(t) - d \log t - 12.\end{aligned}$$

For the polynomial $p(t) := (t^{d+1}) \cdot q(t) \cdot q'(t)$ and for every $t \geq p_0(|x|, |y|)$, we get

$$\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{p(t)}(x \mid r) + \mathrm{pK}^{p(t)}(y_r \mid x, r) - \log p(t),$$

as desired. Finally, consider the case that $k \leq 0$ or $k' \leq 0$. If $k \leq 0$, then $\mathrm{pK}^{q(t)}(x \mid r) \leq \log q(t) + 1$, implying that $\mathrm{pK}^{p(t)}(x \mid r) < \log p(t)$. But then the lemma simply follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(y_r \mid x, r)$. Similarly, if $k' \leq 0$, then $\mathrm{pK}^{p(t)}(y_r \mid x, r) < \log p(t)$, and the lemma follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(x \mid r)$. ◁

3.7 Probabilistic computational depth (original Item1 and its full proof)

The next lemma is a straightforward adaptation of an argument from [11]. We provide a proof for completeness.

Lemma 31 (Computational Depth Upper Bound), Item 1. For all $x \in \{0,1\}^n$ and $r \in \{0,1\}^*$, and for all non-decreasing polynomials p_0 and p, if n is large enough then there exists a time bound t_1 such that $p_0(n) \leq t_1 \leq 2^{n/\log n}$ and

$$\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) \leq O(n/\log n).$$

Proof. We will start by proving Item 1. Given $x \in \{0,1\}^n$ and polynomials p_0 and p, define the polynomial $\tau := p \circ p_0$. For an integer $I \geq 1$, consider the following telescoping sum:

$$\begin{aligned}\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r) = {} & (\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^2(n)}(x \mid r)) \\ & + (\mathrm{pK}^{\tau^2(n)}(x \mid r) - \mathrm{pK}^{\tau^3(n)}(x \mid r)) \\ & + \cdots + (\mathrm{pK}^{\tau^I(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r)),\end{aligned}$$

where $\tau^i(-)$ denotes the composition of τ with itself i times. For any choice of x, p_0, and p as in the statement of the lemma, $\mathrm{pK}^{\tau(n)}(x \mid r) \leq n + d$, for some universal constant $d \geq 0$; hence, the above sum is at most $n + d$. By averaging, there is some index $i_0 \in [I]$ such that

$$\mathrm{pK}^{\tau^{i_0}(n)}(x \mid r) - \mathrm{pK}^{\tau^{i_0+1}(n)}(x \mid r) \leq (n+d)/I.$$

For this i_0, define $t_1 := \tau^{i_0}(n)$. Note that $t_1 \geq \tau(n) \geq p_0(n)$, since $i_0 \geq 1$ and $p(\ell) \geq \ell$ for every input ℓ. Letting $c \in \mathbb{N}$ be such that $\tau(n) \leq n^c$ for sufficiently large n, define $I := \log_c(n/(\log_2 n)^2)$. Then $t_1 \leq n^{c^I} = 2^{n/\log n}$. Moreover,

$$\begin{aligned}\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) &\leq \mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{\tau(t_1)}(x \mid r) \\ &\leq O(n/\log n),\end{aligned}$$

as desired.

4 Probabilistic Worst-Case to Average-Case Reductions (AM case)

4.1 Auxiliary lemmas

Given a language $L \in \mathrm{NP}$ with a corresponding verifier V, and $x \in L$, let y_x be the lexicographically first witness that $x \in L$. It is not hard to compute y_x from x if an NP oracle is available, by performing a standard search-to-decision reduction. For this reason, the time-bounded Kolmogorov complexity of the pair (x, y_x) is essentially that of x, in the presence of an NP oracle. We will show, more generally, that the oracle can be eliminated under an appropriate average-case easiness assumption, even for the following randomized versions of Σ_ℓ^{P}, denoted $\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$.

Definition 32 ($\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$). A language L is in the class $\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$ if there is a $\mathrm{BP} \circ \Sigma_\ell \circ \mathrm{P}$ formula

$$\varphi(x) = \mathrm{BP}_r \exists y_1 \forall y_2 \ldots Q y_\ell \; R(x, y_1, y_2, \ldots, y_\ell, r)$$

for a polytime predicate R and all string variables $y_1, \ldots, y_\ell, r$ of length $\mathrm{poly}(|x|)$, such that, for all $x \in \{0,1\}^n$,

$$x \in L \implies \Pr_r[\exists y_1 \forall y_2 \ldots Q y_\ell \; R(x, y_1, y_2, \ldots, y_\ell, r)] \geq \delta, \text{ and}$$
$$x \notin L \implies \Pr_r[\forall y_1 \exists y_2 \ldots \overline{Q} y_\ell \; \neg R(x, y_1, y_2, \ldots, y_\ell, r)] \geq \delta.$$

Lemma 33 (Oracle Elimination), original Item 1. For any $\ell \geq 1$, suppose $\mathrm{Dist}\Sigma_{\ell+1}^{\mathrm{P}} \subseteq \mathrm{AvgBPP}$. For an arbitrary $L \in \mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$, let $x \in L_n$ be sufficiently large, let $r \in \{0,1\}^{n^c}$ be any random string under the BP quantifier such that there exists a witness for the left-most $\exists$ quantifier in the definition of L, and let $y_{x,r} \in \{0,1\}^{n^c}$ be the lexicographically first such L-witness for x and randomness r, for some constant $c > 0$. There exist polynomials q and q_0 such that, for all sufficiently large $n \in \mathbb{N}$, all $x \in L_n$, and all $t \geq q_0(n)$, we have with probability at least $\delta - 1/10$ over $r \in \{0,1\}^{n^c}$ that an L-witness $y_{x,r}$ exists, and

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t).$$

Moreover, for the case of no BP quantifier, i.e., for $L \in \Sigma_\ell^{\mathrm{P}}$, we get the same conclusions (with probability 1) without conditioning on r.

Proof of Item 1. Define

$$L' := \{(\mathrm{DP}_k(x, y; z), w, w', 1^k, 1^s, 1^n) \mid \exists M \in \{0,1\}^s,$$
$$M^{\Sigma_\ell^{\mathrm{P}}}(w, w') \text{ prints } (x, y) \in \{0,1\}^{n+n^c} \text{ within } |w| \text{ steps, where } |w'| = n^c, \text{ and } s = k - 10\}.$$

More formally, in this and subsequent proofs, the above notation should be interpreted as follows: a string $(u, w, w', 1^k, 1^s, 1^n)$ belongs to L' iff there exist some $(x, y) \in \{0,1\}^{n+n^c}$ and $M \in \{0,1\}^s$ such that $\mathrm{DP}_k(x, y; z) = u$, where $u = (z, \alpha)$ for some $z \in \{0,1\}^{(n+n^c)k}$ and $\alpha \in \{0,1\}^k$, and $M^{\Sigma_\ell^{\mathrm{P}}}(w, w')$ outputs (x, y) within $|w|$ steps, for $|w'| = n^c$. Note that $L' \in \mathrm{NP}^{\Sigma_\ell^{\mathrm{P}}}$. Define a distribution

family $\mathcal{D} = \{D_{\langle (n+n^c)k+k, 2t, n^c, k, s, n\rangle}\}$ as follows: sample $u \sim U_{(n+n^c)k+k}$, $w \sim U_{2t}$, $w' \sim U_{n^c}$, and output $(u, w, w', 1^k, 1^s, 1^n)$, for $s = k - 10$. By assumption, $(L', \mathcal{D}) \in \mathrm{AvgBPP}$. Let B be a randomized algorithm for $(L', \mathcal{D})$ as in Lemma 6. For all (sufficiently large) n, k, and s, and for independent uniformly random u, w, and w',

$$\begin{aligned}
&\Pr_{u,w,w'}[(u, w, w', 1^k, 1^s, 1^n) \in L'] \\
&= \Pr_{u,w,w'}[\exists (x, y) \in \{0,1\}^{n+n^c}, \exists z \in \{0,1\}^{(n+n^c)k}, \exists M \in \{0,1\}^s, \\
&\qquad M^{\Sigma_\ell^{\mathrm{P}}}(w, w') = (x, y) \wedge u = \mathrm{DP}_k(x, y; z)] \\
&\leq \frac{2^{s+|w|+|w'|+|z|}}{2^{k+|w|+|w'|+|z|}} = 2^{s-k} = 2^{-10},
\end{aligned}$$

where the inequality is by a union bound and a counting argument, and the last equality by the condition that $s = k - 10$ in the definition of L'. Hence,

$$\begin{aligned}
\Pr_{u,w,w',r_B}[B(u, w, w', 1^k, 1^s, 1^n) = 1] &\leq \Pr_{u,w,w'}[(u, w, w', 1^k, 1^s, 1^n) \in L'] \\
&\quad + \Pr_{u,w,w',r_B}[B(u, w, w', 1^k, 1^s, 1^n) \neq L'(u, w, w', 1^k, 1^s, 1^n)] \\
&\leq 2^{-10} + (3/n).
\end{aligned}$$

By Markov's inequality, for at least $9/10$ fraction of strings r,

$$\Pr_{u,w,r_B}[B(u, w, r, 1^k, 1^s, 1^n) = 1] \leq 10 \cdot (2^{-10} + (3/n)) \leq 1/10. \tag{18}$$

Let $x \in L_n$ be arbitrary. By definition of the BP_δ quantifier, for at least δ fraction of random strings $r \in \{0,1\}^{n^c}$, there exists an L-witness, and hence, the lexicographically first L-witness $y_{x,r}$. By the above, for at least $\delta - 1/10$ of random r, we have that both a witness $y_{x,r}$ exists and Equation (18) holds. Fix any such r. Observe that for some polynomial q_0 (dependent on L) and some constant d, for any $t \geq q_0(n)$,

$$\mathrm{pK}^{2t, \Sigma_\ell^{\mathrm{P}}}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + d \log n =: s(x, r). \tag{19}$$

In particular, $q_0(n)$ reflects the time required to deterministically compute $y_{x,r}$, given x and r, by a search-to-decision procedure for L using a Σ_ℓ^{P}-oracle. Define $s := s(x, r)$ (which determines $k = s + 10$), and let $t \geq q_0(n)$. By the definitions of $\mathrm{pK}^{2t, \Sigma_\ell^{\mathrm{P}}}$ and L', for every $z \in \{0,1\}^{(n+n^c)k}$,

$$\Pr_w[(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) \in L'] \geq 2/3.$$

Then by the definition of B,

$$\begin{aligned}
&\Pr_{z,w,r_B}[B(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) = 1] \\
&\geq \Pr_{z,w}[(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) \in L'] \cdot \Pr_{r_B}[B(\omega) = L'(\omega) \mid \omega \in L'] \\
&\geq \frac{2}{3}(1 - (1/n)) \geq 2/3 - o(1).
\end{aligned} \tag{20}$$

Comparing Equation (18) with Equation (20), it is clear that $B(-, U_{2t}, r, 1^k, 1^s, 1^n)$ is a randomized algorithm (with advice) that (1/2)-distinguishes $\mathrm{DP}_k(x, y_{x,r}; U_{(n+n^c)k})$ from uniform. Lemma 22 implies that

$$\begin{aligned} \mathrm{pK}^{p'(t)}(x, y_{x,r} \mid r, s, n, t) &\leq k + \log p'(t) \\ &= s + 10 + \log p'(t) \\ &= \mathrm{pK}^t(x \mid r) + d \log n + \log p'(t) + 10, \end{aligned} \tag{21}$$

for some polynomial p' with $p'(t) \geq p_{\mathrm{DP}}(t_B \cdot 3 \cdot (n + n^c))$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, U_{2t}, r, 1^k, 1^s, 1^n)$. As the advice (s, n, t) can be encoded with $\log s + \log n + \log t + O(\log \log n) \leq (3.1) \cdot \log t$ bits, it follows that $\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t)$ for the polynomial $q(t) := t^{d+4} \cdot p'(t)$. The "Moreover" statement follows the same argument, dropping any mention of w' (and r). ◁

Lemma 34 (Witness Compression), original Item 1. For any $\ell \geq 1$, suppose $\mathrm{Dist}\Sigma^{\mathrm{P}}_{\ell+1} \subseteq \mathrm{AvgBPP}$. For an arbitrary $L \in \mathrm{BP}_\delta \circ \Sigma^{\mathrm{P}}_\ell$, let $x \in L_n$ be sufficiently large, $r \in \{0,1\}^{n^c}$ be any good random string, and let $y_{x,r} \in \{0,1\}^{n^c}$ be the lexicographically first L-witness for x with respect to randomness r, for some constant $c > 0$. Let p_0, p and q_0, q be the polynomials from Items 1 of Lemmas 26 and 33 respectively. Then for every $t \geq \max\{p_0(n), q_0(n + n^c)\}$, with probability at least $\delta - 1/5$ over r, an L-witness $y_{x,r}$ exists, and

$$\mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) \leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r)) + 2 \log p(q(t)).$$

Moreover, for $L \in \Sigma^{\mathrm{P}}_\ell$, we get the same conclusions without conditioning on r.

Proof. We start by proving Item 1. By Item 1 of Lemma 33 and Lemma 26, we get by a union bound that, with probability at least $\delta - 2/10$ over r,

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t),$$

and

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) > \mathrm{pK}^{p(q(t))}(x \mid r) + \mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) - \log p(q(t)).$$

Combining the previous two inequalities,

$$\begin{aligned} \mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) &< \mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r) + \log p(q(t)) \\ &\leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r)) + 2 \log p(q(t)). \end{aligned}$$

The "Moreover" part follows by applying the "no random r" versions of Lemma 33 and Lemma 26.

4.2 Case of AM

Theorem 36. If $\mathrm{Dist}\Sigma^{\mathrm{P}}_2 \subseteq \mathrm{AvgBPP}$, then $\mathrm{AM} \subseteq \mathrm{BPTIME}[2^{O(n/\log n)}]$.

Proof. Consider an arbitrary $L \in \mathrm{AM}$ given by a $\mathrm{BP} \circ \Sigma_1^{\mathrm{P}}$ formula such that, for every $x \in \{0,1\}^n$,

$$x \in L \Longrightarrow \mathrm{BP}_{(1-2^{-n})} r \; \exists y \; V(x,y,r),$$
$$x \notin L \Longrightarrow \mathrm{BP}_{(1-2^{-n})} r \; \forall y \; \neg V(x,y,r),$$

for a polytime verifier $V(x,y,r)$, where $|r|, |y| \leq n^c$ for some constant $c > 0$, and the notation $\mathrm{BP}_{(\delta)} r \; \varphi(-,r)$, for a formula φ, means $\Pr_r[\varphi(-,r)] \geq \delta$. For a given (sufficiently large) $x \in L_n$, and a good random string r, let $y_{x,r}$ be the lexicographically first L-witness for x, r. By Lemma 34, for some constant $a > 0$, and for all large enough time bounds t, we have, with probability at least $1 - 2^{-n} - 1/5$ over random strings r, that

$$\mathrm{pK}^{t^a}(y_{x,r} \mid x, r) \leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{t^a}(x \mid r)) + (2a) \log t.$$

By Lemma 31, there exists a time bound $t \leq 2^{O(n/\log n)}$ such that $\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{t^a}(x \mid r) \leq bn/\log n$, for some universal constant $b > 0$. Hence, for some universal constant $d > 0$ (independent of x, r),

$$\mathrm{pK}^{2^{dn/\log n}}(y_{x,r} \mid x, r) \leq dn/\log n.$$

A randomized algorithm for L is now obvious:

- Given $x \in \{0,1\}^n$, sample uniformly random strings $r \in \{0,1\}^{|x|^c}$ and $w \in \{0,1\}^{2^{dn/\log n}}$, and then exhaustively search over all $dn/\log n$-bit machines M, running each $M(w,x,r)$ for $2^{dn/\log n}$ steps to produce a candidate witness y, accepting if $V(x,y,r)$ accepts.

The runtime $2^{O(n/\log n)}$ is clear. For correctness, for every $x \in L_n$, we know there is at least a $(4/5 - 2^{-n})$ fraction of good random strings r for which a witness $y_{x,r}$ will be found by our algorithm, with probability at least $2/3$ over random w's (by the definition of pK). So our algorithm accepts $x \in L$ with probability at least $(2/3)(4/5) - o(1) > 1/2$. On the other hand, for any $x \notin L$, our algorithm may find a witness y and accept x for at most 2^{-n} fraction of r's. ◁

Additional author rKt and pseudodeterministic-PRG context

We denote by $\mathrm{rK}^t_\delta(x)$ the minimum length of a randomized machine M that outputs x with probability at least δ when running for at most t steps (see [38,33]). We simply write $\mathrm{rK}^t(x)$ when $\delta = 2/3$.

Definition 8 (Pseudodeterministic PRG [39,33]). Fix an error function $\varepsilon : \mathbb{N} \to \mathbb{R}$. A function family $G_n : \{0,1\}^\ell \to \{0,1\}^n$ is an ε-error Pseudorandom Generator (PRG) if, for all sufficiently large $n \in \mathbb{N}$, the distribution $G(U_{\ell(n)})$ $\varepsilon(n)$-fools circuits of size n, using the seed length $\ell = \ell(n)$, i.e., for all circuits C of size n on n inputs,

$$\left| \Pr_{r \sim \{0,1\}^n}[C(r) = 1] - \Pr_{\alpha \sim \{0,1\}^{\ell(n)}}[C(G(\alpha)) = 1] \right| \leq \varepsilon.$$

Such a PRG G is called a pseudodeterministic PRG if there is a randomized algorithm M_G such that, for every n and $\alpha \in \{0,1\}^{\ell(n)}$, $M_G(1^n, \alpha)$ outputs the string $G_n(\alpha) \in \{0,1\}^n$ with probability at least $1 - 2^{-n}$ over its internal randomness, and M_G runs in time $2^{O(\ell(n))}$.

Lemma 9 ([19,39]). If BPE contains a language L of circuit complexity $2^{\Omega(n)}$ for almost all input lengths n, then there is an ε-error pseudodeterministic PRG of seed length $O(\log(n/\varepsilon))$.

Lemma 10 ([35, Lemma 2]). $\mathrm{E}^{\Sigma_2^{\mathrm{P}}}$ contains a language of maximum circuit complexity (at least $2^n/n$) for almost all input lengths n. Using padding as in [2], we get the following.

Lemma 11. If $\mathrm{DistP}^{\Sigma_2^{\mathrm{P}}} \subseteq \mathrm{AvgBPP}$, then $\mathrm{E}^{\Sigma_2^{\mathrm{P}}} = \mathrm{BPE}$.

Proof. The inclusion $\mathrm{BPE} \subseteq \mathrm{E}^{\Sigma_2^{\mathrm{P}}}$ follows from the inclusion $\mathrm{BPP} \subseteq \Sigma_2^{\mathrm{P}}$ [22] by padding. For the other direction, let $L \in \mathrm{E}^{\Sigma_2^{\mathrm{P}}}$ be arbitrary. Define its padded version $L_{\mathrm{pad}} = \{(x, 1^{2^{2|x|}}) \mid x \in L\}$. Note that $L_{\mathrm{pad}} \in \mathrm{P}^{\Sigma_2^{\mathrm{P}}}$. Define a family of distributions $D_n = (U_n, 1^{2^{2n}})$, for the uniform distribution over n-bit strings U_n. By assumption, $(L_{\mathrm{pad}}, D_n) \in \mathrm{AvgBPP}$, and so, by Lemma 6, Item 3, there is a randomized polytime algorithm B such that $B(x, 1^{2^{2|x|}}) \neq L(x)$ with probability less than $3 \cdot 2^{-2n} < 1/(3 \cdot 2^n)$, where the probability is over a uniformly random $x \in \{0,1\}^n$ and the internal randomness of B. It follows that, for every $x \in \{0,1\}^n$, $B(x, 1^{2^{2|x|}}) \neq L(x)$ with probability less than $1/3$ over its internal randomness. Hence, the randomized algorithm $B'(x) := B(x, 1^{2^{2|x|}})$ decides if $x \in L$ with probability at least $2/3$, for all sufficiently large inputs $x \in \{0,1\}^n$, yielding $L \in \mathrm{BPE}$. ◁

Corollary 12. If $\mathrm{DistP}^{\Sigma_2^{\mathrm{P}}} \subseteq \mathrm{AvgBPP}$, then there is an ε-error pseudodeterministic PRG of seed length $O(\log(n/\varepsilon))$.

Proof. Immediate by combining Lemmas 9–11. ◁

Lemma 23 (rK^t Reconstruction Lemma). Assume the existence of a pseudodeterministic PRG. For $\varepsilon > 0$, $x \in \{0,1\}^n$, $s \in \mathbb{N}$, and $k \in \mathbb{N}$ satisfying $k \leq 2n$, let D be a randomized algorithm that takes an advice β and runs in time t_D such that D ε-distinguishes $\mathrm{DP}_k(x; U_{nk})$ from U_{nk+k}. Then there is a polynomial p'_{DP} such that

$$\mathrm{rK}^{p'_{\mathrm{DP}}(nt_D/\varepsilon)}(x \mid \beta) \leq k + \log p'_{\mathrm{DP}}(nt_D/\varepsilon).$$

Proof. By averaging, arguing as in Lemma 22 up to Equation (4),

$$\Pr_{r_D \sim \{0,1\}^s}[\mathbb{E}_y[D(\mathrm{DP}_k(x,y), r_D)] - \mathbb{E}_z[D(z, r_D)] \geq \varepsilon/2] \geq \varepsilon/2. \tag{5}$$

That is, with probability at least $\varepsilon/2$ over its internal randomness r_D, $D(-, r_D)$ is an $(\varepsilon/2)$-distinguisher for $\mathrm{DP}_k(x; U_{nk})$ and U_{nk+k}. Let us say that r_D is good

if this is true. Let $\mathrm{Recon}^{(-)} : \{0,1\}^k \times \{0,1\}^r \to \{0,1\}^n$ be the reconstruction procedure from Lemma 14 that works for distinguishing parameter $\varepsilon/2$. By definition of Recon, if r_D is good, then

$$\Pr_{w \sim \{0,1\}^r}[\mathrm{Recon}^{D(-,r_D)}(A(x,w),w) = x] \geq 1/\operatorname{poly}(n/\varepsilon),$$

and so

$$\Pr_{w,r_D}[\mathrm{Recon}^{D(-,r_D)}(A(x,w),w) = x] \geq \frac{\varepsilon}{2}\frac{1}{\operatorname{poly}(n/\varepsilon)} \stackrel{\text{def}}{=} \varepsilon'.$$

Observe that by Lemma 14, the condition $\mathrm{Recon}^{D(-,r_D)}(A(x,w),w) = x$ can be checked by a circuit of size $s = \operatorname{poly}(nt_D/\varepsilon)$ given (r_D, w) as input. Consider an assumed pseudodeterministic PRG

$$G : \{0,1\}^\ell \to \{0,1\}^{|r_D|+|w|}$$

that $(\varepsilon'/2)$-fools all circuits of size s, where $\ell \in O(\log(s/\varepsilon')) \subseteq O(\log(ns/\varepsilon))$. Let M_G be an algorithm as described in Definition 8, running in time $\operatorname{poly}(s/\varepsilon') \leq \operatorname{poly}(nt_D/\varepsilon)$, such that for all $\sigma \in \{0,1\}^\ell$,

$$\Pr_{r_M \sim \{0,1\}^{\operatorname{poly}(nt_D/\varepsilon)}}[M_G(\sigma, r_M) = G(\sigma)] \geq 1 - 2^{-s},$$

where r_M denotes the internal randomness of M_G. By definition of G,

$$\Pr_{\sigma}[\mathrm{Recon}^{D(-,G_0(\sigma))}(A(x,G_1(\sigma)),G_1(\sigma)) = x] \geq \varepsilon'/2,$$

where $G_0(\sigma) = G(\sigma)_{1\ldots|r_D|}$ (the $|r_D|$-length prefix) and $G_1(\sigma) = G(\sigma)_{|r_D|+1\ldots|r_D|+|w|}$ (the remaining suffix). So there must exist a seed $\sigma \in \{0,1\}^\ell$ such that

$$\mathrm{Recon}^{D(-,G_0(\sigma))}(A(x,G_1(\sigma)),G_1(\sigma)) = x.$$

Let σ be a seed with this property, and let $\alpha = A(x, G_1(\sigma)) \in \{0,1\}^k$. The definition of M_G implies that α and σ are such that

$$\Pr_{r_M}[\mathrm{Recon}^{D(-,M^0_G(\sigma,r_M))}(\alpha, M^1_G(\sigma,r_M)) = x] \geq 1 - 2^{-s} > 2/3,$$

where $M^0_G(\sigma, r_M)$ denotes the $|r_D|$-length prefix of $M_G(\sigma, r_M)$, and $M^1_G(\sigma, r_M)$ the remaining suffix of $M_G(\sigma, r_M)$. It is easy to verify that the reconstruction procedure $\mathrm{Recon}^{D(-,M^0_G(\sigma,r_M))}(\alpha, M_G(\sigma,-))$ runs in time $\operatorname{poly}(nt_D/\varepsilon)$ overall. By definition of rK^t, for some polynomial p'_{DP}, we get

$$\begin{aligned}\mathrm{rK}^{p'_{\mathrm{DP}}(nt_D/\varepsilon)}(x \mid \beta) &\leq |\alpha| + |\sigma| + O(\log(ns/\varepsilon)) \\ &\leq k + \log p'_{\mathrm{DP}}(nt_D/\varepsilon),\end{aligned}$$

as required. ◁

Lemma 26, original Item 2. If $\mathrm{DistP}^{\Sigma_2^{\mathrm{P}}} \subseteq \mathrm{AvgBPP}$, then there exist polynomials p' and p'_0 such that for all sufficiently large $n, m \in \mathbb{N}$, all $t \geq p'_0(n,m)$, and all $0 \leq \tau \leq t$ the following holds: for every $x \in \{0,1\}^n$ and every family $\{y_r \in \{0,1\}^m \mid r \in \{0,1\}^\tau\}$, with probability at least 9/10 over $r \in \{0,1\}^\tau$,

$$\mathrm{rK}^t(x, y_r \mid r) > \mathrm{rK}^{p'(t)}(x \mid r) + \mathrm{rK}^{p'(t)}(y_r \mid x, r) - \log p'(t).$$

Moreover, the special case without conditioning on r also holds for both items above, i.e., when r is the empty string (for $\tau = 0$) and y is an arbitrary single string.

Proof of Item 2. Argue as in the proof of Item 1, but at Equation (9), apply Lemma 23 instead of Lemma 22 to get

$$\mathrm{rK}^{q(t)}(x \mid r) \leq k + \log q(t)$$

for some polynomial q such that $q(t) \geq p'_{\mathrm{DP}}(t_B \cdot 3 \cdot n)$, where p'_{DP} is the polynomial from Lemma 23 and t_B denotes the time required to compute $B(-, v, w, r, 1^s)$. We then define $k := \mathrm{rK}^{q(t)}(x \mid r) - \log q(t) - 1$ so the above equation does not hold. Similarly, at Equation (11), apply Lemma 23 to get

$$\mathrm{rK}^{q'(t)}(y_r \mid x, r) \leq k' + \log q'(t)$$

for some polynomial q' with $q'(t) \geq p'_{\mathrm{DP}}(t_{B'} \cdot 8 \cdot m)$, where $t_{B'}$ denotes the time required to compute the distinguisher B' defined as above. Then define $k' := \mathrm{rK}^{q'(t)}(y_r \mid x, r) - \log q'(t) - 1$. Following the proof of Item 1 up to Equation (12), we get that

$$\mathrm{pK}^t(x, y_r \mid r) > k + k' - d \log t - \log n.$$

The definitions of k, k', and rK^t then imply that

$$\begin{aligned}\mathrm{rK}^t(x, y_r \mid r) > \mathrm{rK}^{q(t)}(x \mid r) + \mathrm{rK}^{q'(t)}(y_r \mid x, r) \\ - \log q(t) - \log q'(t) - d \log t - 12,\end{aligned}$$

and so

$$\mathrm{rK}^t(x, y_r \mid r) > \mathrm{rK}^{p(t)}(x \mid r) + \mathrm{rK}^{p(t)}(y_r \mid x, r) - \log p(t)$$

for the polynomial $p(t) := (t^{d+1}) \cdot q(t) \cdot q'(t)$. ◁

Lemma 33, original Item 2. Assume in addition that $\mathrm{DistP}^{\Sigma_2^{\mathrm{P}}} \subseteq \mathrm{AvgBPP}$ (which holds in particular for $\ell \geq 2$). Then there exist polynomials q' and q'_0 such that, for all sufficiently large $n \in \mathbb{N}$, all $x \in L_n$, and all $t \geq q'_0(n)$, we have with probability at least $\delta - 1/10$ over $r \in \{0,1\}^{n^c}$ that an L-witness $y_{x,r}$ exists, and

$$\mathrm{rK}^{q'(t)}(x, y_{x,r} \mid r) \leq \mathrm{rK}^t(x \mid r) + \log q'(t).$$

Moreover, for the case of no BP quantifier, i.e., for $L \in \Sigma_\ell^{\mathrm{P}}$, we get the same conclusions (with probability 1) without conditioning on r.

Proof of Item 2. The proof is similar to that of Item 1, except at Equation (19), we observe that for all $t \geq q_0(n)$,

$$\mathrm{pK}^{2t, \Sigma_\ell^{\mathrm{P}}}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + d \log n \leq \mathrm{rK}^t(x \mid r) + d \log n,$$

and then define $s(x, r) := \mathrm{rK}^t(x \mid r) + d \log n$. We then follow the proof of Item 1 up to Equation (21), where we apply Lemma 23 instead of Lemma 22 to obtain

$$\mathrm{rK}^{p''(t)}(x, y_{x,r} \mid r, s, n, t) \leq \mathrm{rK}^t(x \mid r) + d \log n + \log p''(t) + 10$$

for some polynomial p'' with $p''(t) \geq p'_{\mathrm{DP}}(t_B \cdot 3 \cdot (n + n^c))$, where p'_{DP} is the polynomial from Lemma 23 and t_B denotes the time required to compute $B(-, U_{2t}, r, 1^k, 1^s, 1^n)$. It follows that $\mathrm{rK}^{q'(t)}(x, y_{x,r} \mid r) \leq \mathrm{rK}^t(x \mid r) + \log q'(t)$ for the polynomial $q'(t) := t^{d+4} \cdot p''(t)$. ◁

Lemma 34, original Item 2. Assume in addition that $\mathrm{DistP}^{\Sigma_2^{\mathrm{P}}} \subseteq \mathrm{AvgBPP}$ (which holds in particular when $\ell \geq 2$). Let p'_0, p' and q'_0, q' be the polynomials from Items 2 of Lemmas 26 and 33 respectively. Then for every $t \geq \max\{p'_0(n), q'_0(n+n^c)\}$, with probability at least $\delta - 1/5$ over r, an L-witness $y_{x,r}$ exists, and

$$\mathrm{rK}^{p'(q'(t))}(y_{x,r} \mid x, r) \leq (\mathrm{rK}^t(x \mid r) - \mathrm{rK}^{p'(q'(t))}(x \mid r)) + 2\log p'(q'(t)).$$

Moreover, for $L \in \Sigma_\ell^{\mathrm{P}}$, we get the same conclusions without conditioning on r. The proof of Item 2 is the same, except we apply Items 2 of Lemmas 33 and 26 respectively.

4.4 Case of PH (main promise-PH simulation)

It is observed in [7, Corollary 17] that, for every $\ell \geq 1$, $\Sigma_\ell^{\mathrm{P}} = \mathrm{lin}\,\Sigma_\ell^{\mathrm{P}}$, under the assumption that $\mathrm{Dist}\Sigma_{\ell+1}^{\mathrm{P}} \subseteq \mathrm{AvgP}$. We will show a similar result for the class promise-$\Sigma_\ell^{\mathrm{BP}}$ defined below.

Definition 39 (promise-$\Sigma_\ell^{\mathrm{BP}}$ and promise-$\mathrm{lin}\,\Sigma_\ell^{\mathrm{BP}}$). A promise problem $\Pi = (\Pi_Y, \Pi_N) \in$ promise-$\Sigma_\ell^{\mathrm{BP}}$ for $\ell \geq 0$, if there is a $\Sigma_\ell \circ \mathrm{BP} \circ \mathrm{P}$ formula

$$\varphi(x) = \exists y_1 \forall y_2 \ldots Q y_\ell \ \mathrm{BP}_r \ R(x, y_1, y_2, \ldots, y_\ell, r),$$

for a polytime predicate R and all string variables $y_1, \ldots, y_\ell, r$ of length polynomial in $|x|$, such that, for all $x \in \{0,1\}^n$,

$$\begin{aligned} x \in \Pi_Y &\Longrightarrow \exists y_1 \forall y_2 \ldots Q y_\ell \ \mathrm{BP}_{(2/3)r} \ R(x, y_1, y_2, \ldots, y_\ell, r), \text{ and} \\ x \in \Pi_N &\Longrightarrow \forall y_1 \exists y_2 \ldots \overline{Q} y_\ell \ \mathrm{BP}_{(2/3)r} \ \neg R(x, y_1, y_2, \ldots, y_\ell, r), \end{aligned}$$

where $\mathrm{BP}_{(\delta)r} \ R(-, r)$ means that $\Pr_r[R(-, r)] \geq \delta$. The constant $2/3$ in the definition above can be changed to $1 - 2^{-\mathrm{poly}(n)}$ by standard success amplification techniques (even if we start with any pair $\delta_Y > 1 - \delta_N + \varepsilon$ of thresholds, for some $\varepsilon \geq 1/\mathrm{poly}(n)$, where δ_Y is the probability $R(-, r)$ accepts, and δ_N the probability $\neg R(-, r)$ accepts). The promise class promise-$\mathrm{lin}\,\Sigma_\ell^{\mathrm{BP}}$ is defined the same way as above, with an extra condition that, for some constant $a > 0$, $|y_i| \leq an$ for all $1 \leq i \leq \ell$ (but the randomness r may still be of any $\mathrm{poly}(n)$ length).

Theorem 40 (Witness Compression for PH). For any $\ell \geq 0$, suppose $\mathrm{Dist}\Sigma_{\ell+2}^{\mathrm{P}} \subseteq \mathrm{AvgBPP}$. Then

$$\text{promise-}\Sigma_\ell^{\mathrm{BP}} = \text{promise-}\mathrm{lin}\,\Sigma_\ell^{\mathrm{BP}}.$$

Proof. The proof is by induction on ℓ. The base case of $\ell = 0$ is trivially true. For any $\ell > 0$, suppose the claim is true for $\ell - 1$. Consider an arbitrary promise problem $\Pi = (\Pi_Y, \Pi_N) \in$ promise-$\Sigma_\ell^{\mathrm{BP}}$. That is, for every $x \in \{0,1\}^n$,

$$\begin{aligned} x \in \Pi_Y &\Longrightarrow \exists y \ \Pi_{\ell-1} z \ \mathrm{BP}_{(1-2^{-n})r} \ R(x, y, z, r), \\ x \in \Pi_N &\Longrightarrow \forall y \ \Sigma_{\ell-1} z \ \mathrm{BP}_{(1-2^{-n})r} \ \neg R(x, y, z, r), \end{aligned}$$

for some polytime predicate R, where z is a sequence of $\ell-1$ strings $z_1, \ldots, z_{\ell-1}$ under the corresponding $\ell-1$ quantifiers of the $\Pi_{\ell-1}$ pattern, and $|y|, |r|, |z_i| \leq q(n)$, for some polynomial q, for all $1 \leq i \leq \ell-1$. As in the proof of the inclusion $\mathrm{BPP} \subseteq \Sigma_2^{\mathrm{P}} \cap \Pi_2^{\mathrm{P}}$ [22], we replace the BP quantifier by the $\exists\forall$ pattern if ℓ is odd, or by the $\forall\exists$ pattern if ℓ is even. We get that there exists a polytime predicate R' such that, for every $x \in \{0,1\}^n$,

$$x \in \Pi_Y \Longrightarrow \exists y\ \Pi_{\ell-1} z\ Q_0 z'\ Q_1 z''\ R'(x, y, z, z', z''),$$
$$x \in \Pi_N \Longrightarrow \forall y\ \Sigma_{\ell-1} z\ Q_1 z'\ Q_0 z''\ \neg R'(x, y, z, z', z''),$$

where z', z'' have the lengths polynomial in n and $Q_0 Q_1 = \exists\forall$ if ℓ is odd, and $Q_0 Q_1 = \forall\exists$ if ℓ is even. Note that the above transformation shows that our promise problem $\Pi \in \text{promise-}\Sigma_{\ell+1}^{\mathrm{P}}$. For any given $x \in \Pi_Y$, let y_x be the lexicographically first string y such that

$$\Pi_{\ell-1} z\ Q_0 z'\ Q_1 z''\ R'(x, y, z, z', z''). \tag{23}$$

Claim 41. For x and y_x as above,

$$\Pi_{\ell-1} z\ \mathrm{BP}_{(1/\,\mathrm{poly}(n))} r\ R(x, y_x, z, r).$$

Proof. Immediate from the properties of Lautemann's proof of $\mathrm{BPP} \subseteq \Sigma_2^{\mathrm{P}}$ [22]. Recall that in that proof, the subformula $\mathrm{BP}_r\ R(-, r)$ is replaced by

$$\exists u_1 \ldots u_k\ \forall r\ \bigvee_{i=1}^{k} R(-, r \oplus u_i),$$

for some $k \leq \mathrm{poly}(|r|)$, where $|u_i| = |r|$, for all $1 \leq i \leq k$, and $\oplus$ is the bit-wise XOR. If $\Pr_r[R(-, r)] < 1/k$, then such a collection of $u_1, \ldots, u_k$ cannot exist. Indeed, each "shifted" predicate $R(-, r \oplus u_i)$, for $1 \leq i \leq k$, accepts less than $1/k$ fraction of $r \in \{0,1\}^{|r|}$, and hence, by the union bound, the OR of any k such shifts accepts less than $k \cdot (1/k) = 1$ fraction of $r \in \{0,1\}^{|r|}$. $\lhd$ On the other hand, for every $x \in \Pi_N$ and every y, we have

$$\Sigma_{\ell-1} z\ \mathrm{BP}_{(1-2^{-n})} r\ \neg R(x, y, z, r).$$

Using Claim 41, by standard success amplification, for a random string r' of length $q'(n)$, for some polynomial q', and a polytime predicate R^{boost}, we get for every $x \in \{0,1\}^n$ and y_x as defined above,

$$x \in \Pi_Y \Longrightarrow \Pi_{\ell-1} z\ \mathrm{BP}_{(1-2^{-n})} r'\ R^{\mathrm{boost}}(x, y_x, z, r'), \tag{24}$$
$$x \in \Pi_N \Longrightarrow \forall y\ \Sigma_{\ell-1} z\ \mathrm{BP}_{(1-2^{-n})} r'\ \neg R^{\mathrm{boost}}(x, y, z, r'). \tag{25}$$

Claim 42. For some universal constant $a > 0$ and some polynomial p', for every n-bit $x \in \Pi_Y$ with a witness y_x as defined above, we have

$$\mathrm{rK}^{p'(n)}(y_x \mid x) \leq an.$$

Proof. Using Equation (23), we get by Item 2 of Lemma 34 (since $\mathrm{Dist}\Sigma_3^{\mathrm{P}} \subseteq \mathrm{AvgBPP}$), that, for some polynomial p and all large enough t,

$$\mathrm{rK}^{p(t)}(y_x \mid x) \leq \mathrm{rK}^t(x) - \mathrm{rK}^{p(t)}(x) + 2\log p(t).$$

Since, for any t, $\mathrm{rK}^t(x) \leq O(n)$, the claim follows. $\triangleleft$ Consider the formula

$$\varphi(x) = \exists y' \ (|y'| \leq an)\ \Pi_{\ell-1} z\ \mathrm{BP}_w\ R''(x, y', z, w), \tag{26}$$

where the predicate R'' is computed by the following polytime algorithm A:

- On input x, y', z, and a string $w = w'r'$, where w' is of length $p'(n)$, and r' is of length $q'(n)$, first run the universal machine $U(y', x, w')$ for $p'(n)$ steps, getting an output y. Then run $R^{\mathrm{boost}}(x, y, z, r')$, accepting iff R^{boost} accepts.

We claim that Equation (26) is a correct $\Sigma_\ell^{\mathrm{BP}}$ formula for the promise problem Π. Indeed, for any $x \in \Pi_Y$, by Claim 42, there is a string y' that is decompressed to y_x with high probability over random strings w' (say, with probability at least $3/4$). By Equation (24), $R^{\mathrm{boost}}(x, U(y', x, w'), z, r')$ accepts with probability at least $(3/4)(1 - 2^{-n}) > 2/3$ over strings $w = w'r'$. On the other hand, for any $x \in \Pi_N$, Equation (25) implies that

$$\forall y'\ \Sigma_{\ell-1} z\ \mathrm{BP}_{(1-2^{-n})} w\ \neg R''(x, y', z, w).$$

Next, using Equation (26), define a new promise problem $\Pi' = (\Pi'_Y, \Pi'_N) \in \text{promise-}\Pi_{\ell-1}^{\mathrm{BP}}$:

$$\Pi'_Y = \{(x, y') \mid |y'| \leq a|x|,\ \Pi_{\ell-1} z\ \mathrm{BP}_w\ R''(x, y', z, w)\},$$
$$\Pi'_N = \{(x, y') \mid |y'| \leq a|x|,\ \Sigma_{\ell-1} z\ \mathrm{BP}_w\ \neg R''(x, y', z, w)\}.$$

By the Inductive Hypothesis, $\Pi' \in \text{promise-}\mathrm{lin}\,\Pi_{\ell-1}^{\mathrm{BP}}$, with all quantified binary strings of length $O(|x| + |y'|) \leq O(n)$. It follows that $\Pi \in \text{promise-}\mathrm{lin}\,\Sigma_\ell^{\mathrm{BP}}$. $\triangleleft$

Theorem 43. For any $\ell \geq 0$,

$$\mathrm{Dist}\Sigma_{\ell+2}^{\mathrm{P}} \subseteq \mathrm{AvgBPP} \Longrightarrow \text{promise-}\Sigma_\ell^{\mathrm{P}} \subseteq \text{promise-}\mathrm{BPTIME}[2^{O(n/\log n)}].$$

Proof. The proof is by induction on ℓ. The base case of $\ell = 0$ is trivially true. Consider any $\ell \geq 1$, and assume the claim for $\ell - 1$. Let $\Pi = (\Pi_Y, \Pi_N) \in \text{promise-}\Sigma_\ell^{\mathrm{P}}$ be arbitrary. By Theorem 40, $\Pi \in \text{promise-}\mathrm{lin}\,\Sigma_\ell^{\mathrm{BP}}$ via some formula

$$\varphi(x) = \exists y\ \Pi_{\ell-1} z\ \mathrm{BP}_r\ R(x, y, z, r),$$

with a polytime R, linearly-bounded y and $z = (z_1, \ldots, z_{\ell-1})$ (under the $\Pi_{\ell-1}$ quantifiers), and a poly-bounded r. Arguing as in the proof of Theorem 40 (see the proof of Claim 42), but using Lemma 31 (for the case of pK) to bound the computational depth of an input x by $O(n/\log n)$ with a time bound $t \leq 2^{O(n/\log n)}$, we get that, for some universal constant $a > 0$, every $x \in \Pi_Y$ has a witness y_x with

$$\mathrm{pK}^{2^{an/\log n}}(y_x \mid x) \leq an/\log n. \tag{27}$$

Applying success amplification to R, we get a new polytime predicate R^{boost} with poly-bounded randomness r' such that we get for every $x \in \{0,1\}^n$ and y_x as defined above,

$$x \in \Pi_Y \implies \Pi_{\ell-1} z \ \text{BP}_{(1-2^{-n})} r' \ R^{\text{boost}}(x, y_x, z, r'), \tag{28}$$

$$x \in \Pi_N \implies \forall y \ \Sigma_{\ell-1} z \ \text{BP}_{(1-2^{-n})} r' \ \neg R^{\text{boost}}(x, y, z, r'). \tag{29}$$

Define a promise problem $\Pi' = (\Pi'_Y, \Pi'_N) \in \text{promise-}\Pi^{\text{BP}}_{\ell-1}$:

$$\Pi'_Y = \{(x,y) \mid \Pi_{\ell-1} z \ \text{BP}_{r'} \ R^{\text{boost}}(x,y,z,r')\},$$
$$\Pi'_N = \{(x,y) \mid \Sigma_{\ell-1} z \ \text{BP}_{r'} \ \neg R^{\text{boost}}(x,y,z,r')\}.$$

By the Inductive Hypothesis, $\Pi' \in \text{promise-BPTIME}[2^{O(n/\log n)}]$, where $|x| = n$, via some randomized algorithm A, with error probability at most 2^{-n}. We now solve Π with the following randomized algorithm B:

- Given $x \in \{0,1\}^n$, choose a random string w of length $2^{an/\log n}$, and then enumerate over all machines $M \in \{0,1\}^{an/\log n}$, and run $M(w,x)$ for $2^{an/\log n}$ steps, getting a candidate witness y of length $O(n)$. Run $A(x,y)$, accepting iff A accepts.

The time complexity of the described algorithm B is clearly $2^{O(n/\log n)}$. For correctness, for every $x \in \Pi_Y$, with probability at least $3/4$ over w, our algorithm will consider a short description of a TM M such that $M(w,x) = y_x$. Since $(x, y_x) \in \Pi'_Y$, we get that $A(x, y_x)$ will accept with probability at least $1 - 2^{-n}$ over its internal randomness. Hence, the algorithm B accepts x with probability at least $(3/4)(1-2^{-n}) > 2/3$. On the other hand, for every $x \in \Pi_N$, for every w and all M, the string $y = M(w,x)$ is such that $(x,y) \in \Pi'_N$, and so $A(x,y)$ accepts with probability at most 2^{-n}. It follows that the probability that $B(x)$ accepts in this case is

$$\mathbb{E}_{w,A}[\exists v \in \{0,1\}^{an/\log n} : A(x, U(v,x,w))] \leq 2^{an/\log n} \cdot \mathbb{E}_w(2^{-n}) \leq 2^{-n/2},$$

for all sufficiently large n. ◁

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